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 Studierichting: Informatica
 Oefeningenreeks 2D nr. 10.7

$$\lim_{x \rightarrow -2\sqrt{5}} \frac{\sqrt{5}x^2 + 9x - 2\sqrt{5}}{2x^2 + 3x\sqrt{5} - 10} = \frac{0}{0}$$

$$\sqrt{5}x^2 + 9x - 2\sqrt{5}:$$

$$\frac{-2\sqrt{5} \left| \begin{array}{cc|c} \sqrt{5} & 9 & -2\sqrt{5} \\ & -10 & 2\sqrt{5} \end{array} \right|}{\left| \begin{array}{cc|c} \sqrt{5} & -1 & 0 \end{array} \right|} \Rightarrow (x + 2\sqrt{5})(\sqrt{5}x - 1)$$

$$2x^2 + 3x\sqrt{5} - 10:$$

$$\frac{-2\sqrt{5} \left| \begin{array}{cc|c} 2 & 3\sqrt{5} & -10 \\ & -4\sqrt{5} & 10 \end{array} \right|}{\left| \begin{array}{cc|c} 2 & -\sqrt{5} & 0 \end{array} \right|} \Rightarrow (x + 2\sqrt{5})(2x - \sqrt{5})$$

$$\Rightarrow \lim_{x \rightarrow -2\sqrt{5}} \frac{(x + 2\sqrt{5})(\sqrt{5}x - 1)}{(x + 2\sqrt{5})(2x - \sqrt{5})}$$

$$= \lim_{x \rightarrow -2\sqrt{5}} \frac{(\sqrt{5}x - 1)}{(2x - \sqrt{5})}$$

$$= \frac{11}{5\sqrt{5}}$$

References